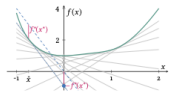
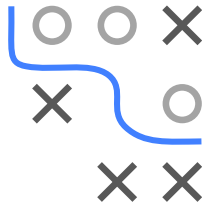


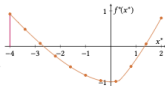
# Optimization in Machine Learning

## Constrained Optimization

## Other forms of duality



(a) The function  $f$  (solid) and the linear function of  $x^* = -4$  (dashed) and its shifted tangent (dotted).



(b) The Fenchel conjugate  $f^*$  of  $f$ .

## Learning goals

- Dual norms
- Conjugate functions
- Fenchel duality
- Examples in statistics

# CONSTRAINED MINIMIZATION AND DUAL NORMS

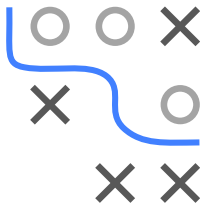
- Consider norm minimization under linear constraints in its primal form:

$$\min_{\mathbf{x} \in \mathbb{R}^d} \|\mathbf{x}\| \quad \text{s.t. } \mathbf{G}\mathbf{x} = \mathbf{h}$$

- $\|\cdot\|$  is some norm function
- If the norm is the  $L_1$  norm, this is the famous basis pursuit problem

► [Click for source](#)

- Question:** Is there a more straightforward way to solve constrained optimization problems involving norms?



# CONSTRAINED MINIMIZATION AND DUAL NORMS

- The concept of the **dual norm** from functional analysis can help

**Definition** Let  $\|\mathbf{x}\|$  be the norm of  $\mathbf{x}$ . The dual norm  $\|\mathbf{x}\|_*$  is defined as

$$\|\mathbf{x}\|_* = \max_{\|\mathbf{z}\| \leq 1} \mathbf{z}^T \mathbf{x}$$

- If  $\|\mathbf{x}\|$  is a norm and  $\|\mathbf{x}\|_*$  its dual norm, then

$$|\mathbf{z}^T \mathbf{x}| \leq \|\mathbf{z}\| \|\mathbf{x}\|_*$$

**Example** The dual norm of the  $L_p$  norm  $\|\cdot\|_p$  is the  $L_q$  norm  $\|\cdot\|_q$  where  $1/p + 1/q = 1$

